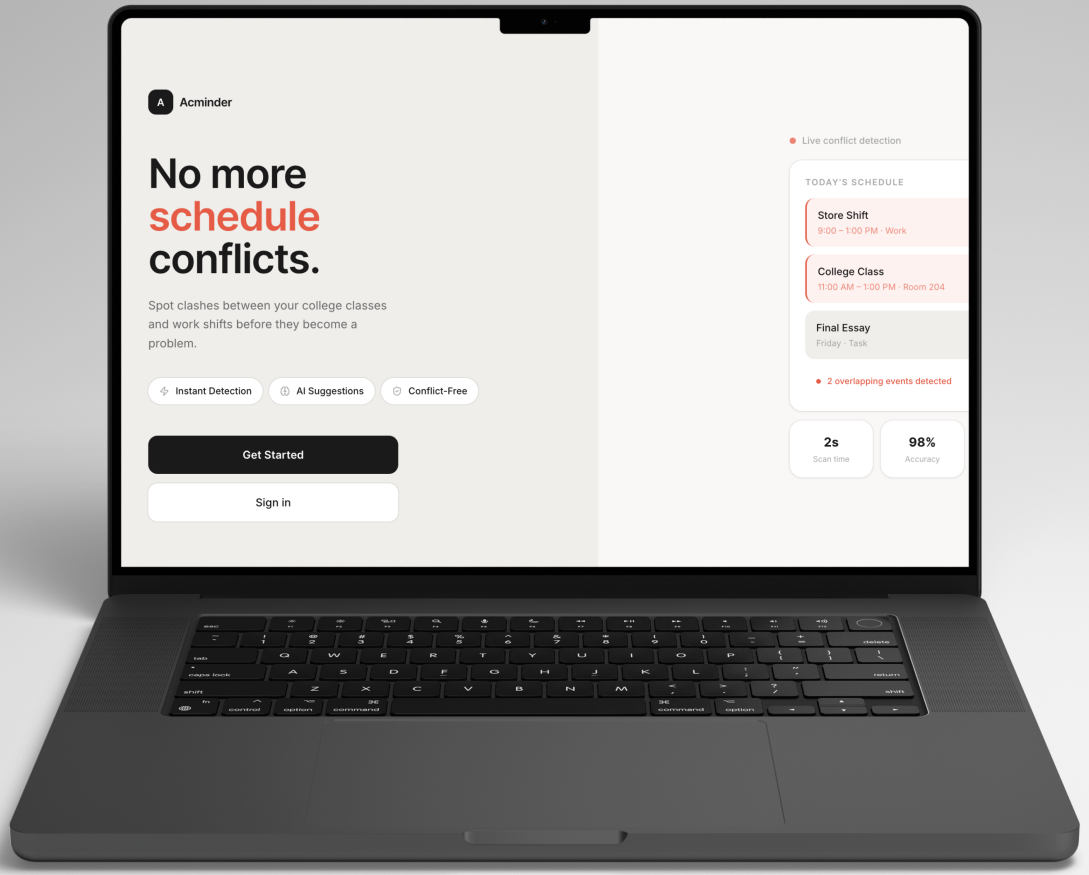
Acminder

Proactive Schedule Management for International Students

CAPSTONE PRESENTATION

Mohammad Hilal Malik


A
Acminder
No more
schedule
conflicts.
Spot clashes between your college classes and work shifts before they become a problem.
Instant Detection
AI Suggestions
Conflict-Free
Get Started
Sign in
Live conflict detection
TODAY'S SCHEDULE
Store Shift
9:00 – 1:00 PM · Work
College Class
11:00 AM – 1:00 PM · Room 204
Final Essay
Friday · Task
2 overlapping events detected
2s
Scan time
98%
Accuracy

Overview: The Acminder Vision

My Vision

A dedicated proactive assistant for international students in Canada, designed to bridge the gap between rigorous academic success and complex work-life balance requirements.

Core Mission

To eliminate scheduling conflicts and reduce the cognitive burden of managing student visa compliance through intelligent automation and unified schedule tracking.

Problem Statement: The Conflict

The Conflict

Balancing strict **20-hour work limits** with full-time academic loads and visa compliance is a constant, high-stakes struggle for international students.

The Consequences

Extreme stress levels reported by **86%** of students, with **43%** missing or almost missing deadlines in just the last 4 weeks.

Target User: The International Student Persona

The Persona

Full-time international students in Canada, often managing high financial stakes and navigating a new academic system.

The Context

Working 20+ hours per week while maintaining a full course load and managing Moodle/LMS deadlines.

The Constraints

Strict visa compliance requirements and high-stakes academic performance goals with zero margin for error.

Process Flow & Design Iteration

PHASE 01

Discovery

Market research and user surveys to identify the "Double Burden" conflict and validate the core problem space.

PHASE 02

Wireframing

Developing Lo-Fi wireframes to establish information hierarchy and intuitive navigation for the dashboard.

PHASE 03

Prototyping

Building a Hi-Fi coded prototype using React and TypeScript to test real-time conflict detection logic.

PHASE 04

Testing

Conducting user testing to identify friction points, followed by iterative fixes in the final coded solution.

Discovery & Market Research Insights

100%

Employment Rate

Every international student surveyed works during the semester to support their living costs.

57%

High Workload

Over half of the participants work 20+ hours per week, pushing the limits of visa regulations.

100%

Conflict Demand

Participants specifically requested automated alerts for overlaps between work and deadlines.

Comparative Analysis: Competitive Edge

Feature	Google/Apple Cal	Moodle / LMS	Acminder
Unified Dashboard	Manual Entry Only	Academic Only	Automated Sync
Conflict Detection	Visual Only	None	Proactive Alerts
Nudge System	Standard Alerts	Email Only	High-Priority Voice
Visa Compliance	No Context	No Context	Work-Hour Tracking

MoSCoW: Roadmap & Future Scope

MUST HAVE

Moodle / iCal Import

Conflict Detection

Nudge Alerts

Unified Dashboard

SHOULD HAVE

Stress Forecasting

Resolution Suggestions

Manual Shift Entry

COULD HAVE

AI Study Planning

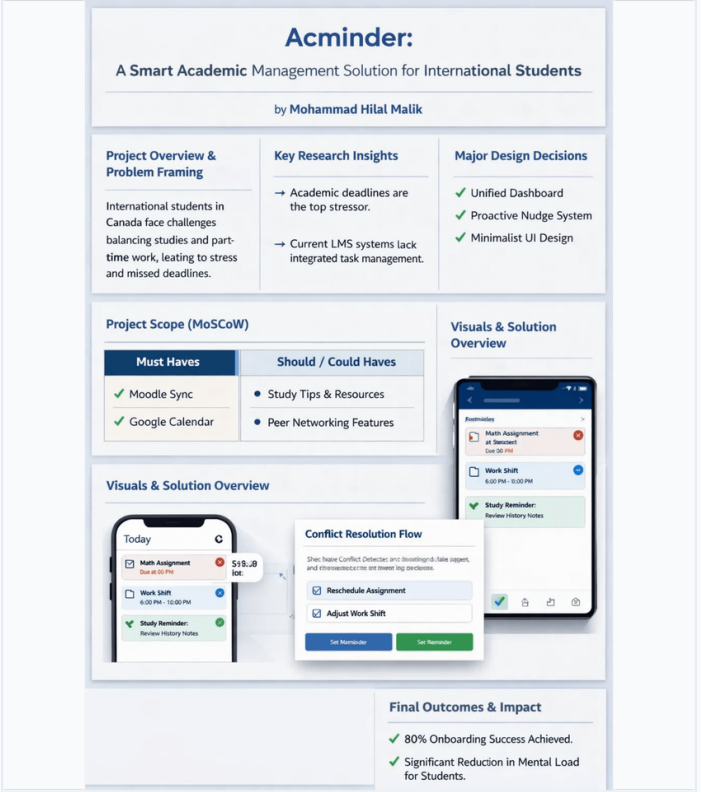
Institutional Scaling

Dark Mode Theme

WON'T HAVE

Social Integration

Marketplace Features



Lo-Fi Wireframes: Structural Foundation

Dashboard Hierarchy

Prioritizing the "Today-First" view. Early sketches focused on making the conflict banner the most prominent element to ensure immediate user action.

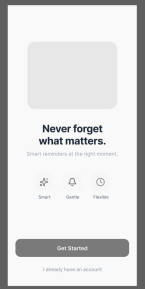
Conflict Resolution

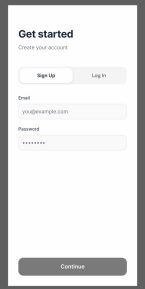
Explored a modal-driven flow to minimize context switching. The focus was on presenting binary choices for rapid decision-making during busy student schedules.

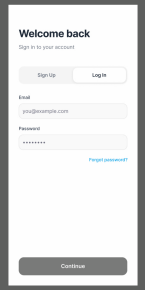
Onboarding Logic

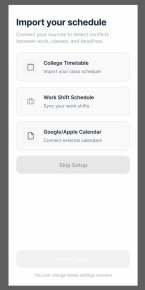
A step-by-step wireframe sequence designed to simplify the multi-source sync process, ensuring users could connect Moodle and Work calendars in under 3 minutes.

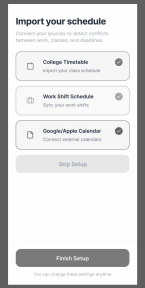
Lo-Fi Wireframes: Structural Foundation

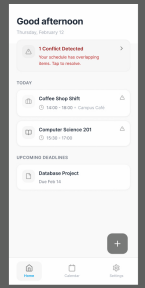












Interaction & Behaviour:

- Users enter their email and a new password.
- The Continue button becomes active upon valid input.
- Tapping Log In on the toggle switches to the Log In screen.

Accessibility Considerations: Input fields have clear labels and sufficient touch targets. Password masking is standard, but a toggle to reveal the password (not shown here) would enhance usability and accessibility

Interaction & Behaviour:

- Tapping any of the integration option card will open a modal to connect needed calendars.
- The Finish Setup button remains disabled until at least one integration source is selected or the Skip Setup button is tapped.
- Tapping Skip Setup proceeds to the Home screen without importing schedules.

Accessibility Considerations:
Large, clearly defined cards for integration options ensure easy tapping. The disabled state of the Finish Setup button provides visual feedback.

Interaction & Behaviour:

- Tapping the Conflict Alert Banner banner navigates to the Conflict Detail Page .
- Tapping on individual scheduled items (e.g., Coffee Shop Shift) would navigate to their respective Item Detail Page .
- Tapping the FAB would navigate to the Add New Item screen

Accessibility Considerations:
Large, clearly defined cards for integration options ensure easy tapping. The disabled state of the Finish Setup button provides visual feedback.

Interaction & Behaviour:
Tapping the Get Started button navigates the user to the Sign Up screen.
Tapping the I already have an account link navigates the user to the Log In screen.

Accessibility Considerations:
The contrast ratio between the text and background is high, ensuring readability. The primary call-to-action is clearly distinguishable

Interaction & Behaviour:

- Users enter their registered email and password.
- The Continue button becomes active upon valid input. Tapping Sign Up on the toggle switches to the Sign Up screen.
- Tapping Forgot password? navigates to a password recovery flow (not shown).

Accessibility Considerations: Clear labels and sufficient touch targets for input fields. The Forgot password? link is clearly visible and clickable

Interaction & Behaviour:

- Tapping the Finish Setup button navigates the user to the Home screen, presumably with their imported schedules now integrated.
- Users can deselect options if they change their mind before finishing setup.

Accessibility Considerations:
The check mark provides clear visual confirmation of selection. The enabled state of the Finish Setup button is distinct from its disabled state

Tech Stack: Technical Foundation

Core Frameworks

React 18 + Vite

Leveraging a modern, component-based architecture for a fast and responsive Single Page Application (SPA) experience.

Language & Safety

TypeScript

Ensuring robust code quality and type safety, which is critical for handling complex schedule data and conflict logic.

UI & Routing

Tailwind CSS React Router v7

Utilizing utility-first styling for rapid UI development and advanced routing for seamless, app-like navigation.

Hi-Fi Coded Prototype: Functionality

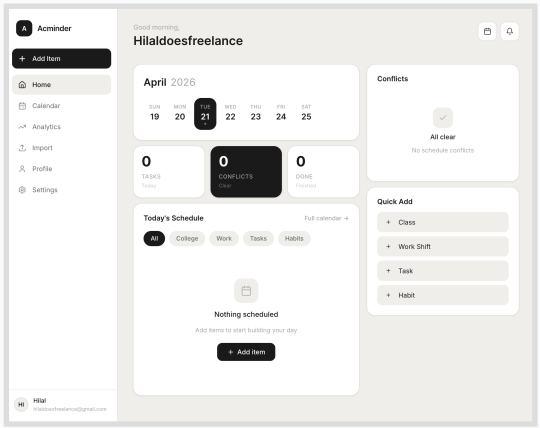
Core Functionality

A fully functional **React application** featuring a responsive, state-driven UI. It includes a unified dashboard, real-time conflict detection using a robust conflict detection engine, and a proactive nudge system for critical alerts with in app notifications.

Technical Implementation

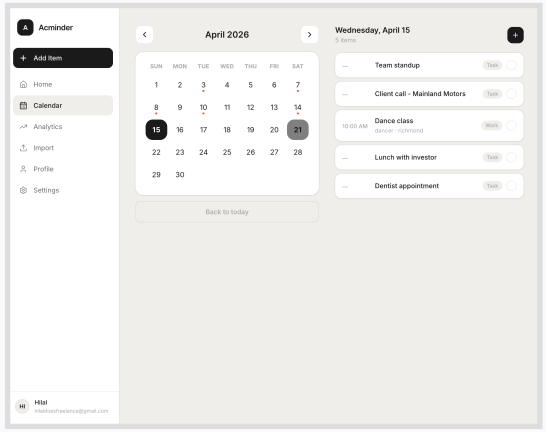
Built with **TypeScript** for type safety and **Vite** for optimized performance. The application uses a modular component architecture, ensuring scalability and professional-grade code quality.

Interface Showcase: Acminder UI



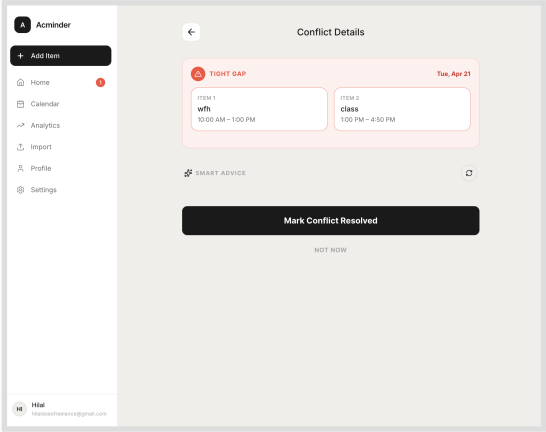
Unified Dashboard

Color-coded daily view with conflict alerts.



Calendar View

Monthly overview of shifts and deadlines.



Conflict Resolution

Smart suggestions for schedule overlaps.

User Testing: Critical Findings

Functionality & Navigation

- **Sign-Up Block:** Generic errors during rate limits with no user instructions.
- **Calendar Friction:** Empty dates are non-interactive for adding items.
- **Missing Mobile View:** Hourly timeline hidden on mobile devices.
- **Accidental Logout:** No confirmation pop-up on the logout button.

Content & Terminology

- **Naming Confusion:** Inconsistent terms for "Work Shift" across screens.
- **Inconsistent Placeholders:** Form examples don't match categories.
- **Vague Conflicts:** Shows overlap time but not the specific duration.
- **Navigation Traps:** Missing "Back" button on the Import screen.

Design & Accessibility

- **Tiny Text:** 8px-10px labels fail mobile accessibility standards.
- **Invisible Click Areas:** Date picker shadows other input buttons.
- **False Status:** Sync says "Live" before account connection.
- **Keyboard Issues:** No highlight rings for selected buttons.

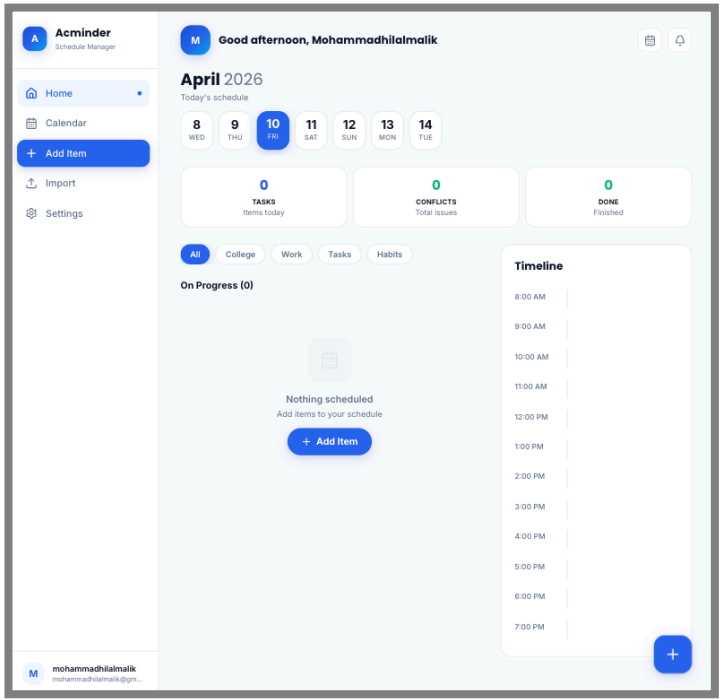
Solutions: Post-Testing Iterations

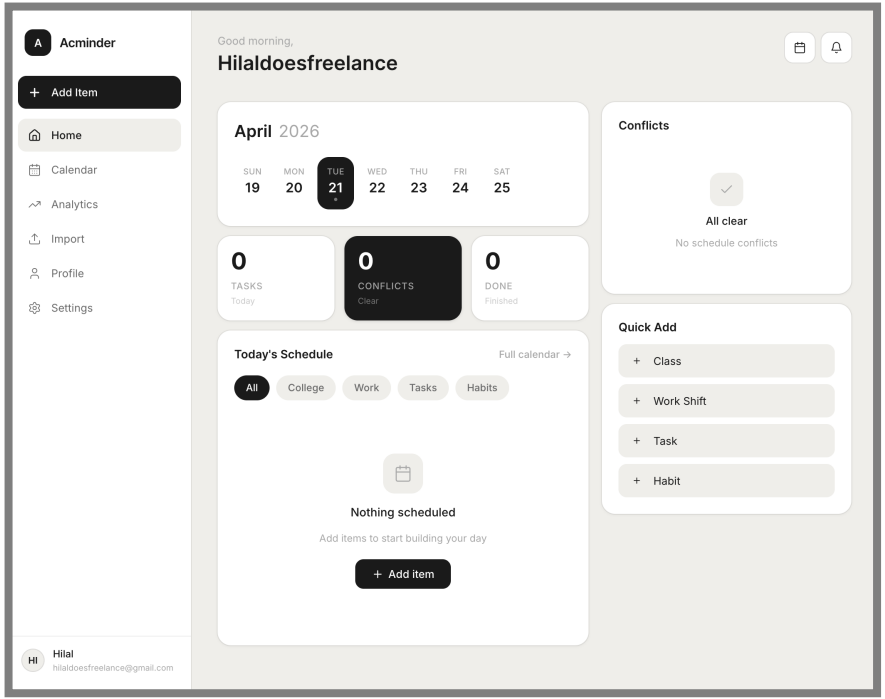
UX & Navigation Fixes

- ✓ Confirmed Logout : Added a confirmation dialog to prevent accidental session termination.
- ✓ Mobile Timeline : Restored visibility of the hourly breakdown for mobile users.
- ✓ Import Navigation : Added "Back" buttons to the onboarding flow for better user control.

Content & Accessibility

- ✓ Standardized Terms : Unified terminology across all screens (e.g., "Work Shift" vs "Shift").
- ✓ Typography Update : Increased minimum font sizes to 12px+ to meet accessibility standards.
- ✓ Conflict Clarity : Added explicit duration labels (e.g., "1h Overlap") to conflict alerts.





Updated user interface after user testing

The Future of Acminder

01 Direct LMS API Integration

Moving beyond one-way iCal feeds to full two-way synchronization with Moodle, Canvas, and Blackboard for real-time assignment tracking and status updates.

02 AI-Driven Weekly Planning

Developing an intelligent scheduling engine that suggests the most efficient study blocks based on a student's energy levels, work shifts, and upcoming deadline intensity.

03 Institutional Scaling

Expanding Acminder's support to more Canadian post-secondary institutions, helping a broader community of international students succeed through institutional partnerships.

The Future of Acminder

01 People reaching out on LinkedIn to collaborate.



Live Demo: Experience Acminder

acminder.netlify.app



Scan the QR code on your poster or visit the URL above to explore the unified dashboard and real-time conflict resolution flow.